

2023
Higher School Certificate
Trial Examination

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your student name and/or number at the top of every page

Total marks – 100

Section I – 10 marks (pages 3 - 5)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 6 - 22)

- Attempt Questions 11 – 33
- Allow about 2 hours and 45 minutes for this section

This paper MUST NOT be removed from the examination room.

STUDENT NAME/NUMBER.....

Section I**10 Marks****Attempt Questions 1-10.****Allow about 15 minutes for this section.**

Select the alternative A, B, C, D that best answers the question and indicate your choice with a cross (X) in the appropriate space on the grid below.

	A	B	C	D
1				
2				
3				
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9				
10				

Section I

10 Marks

Attempt Questions 1-10.

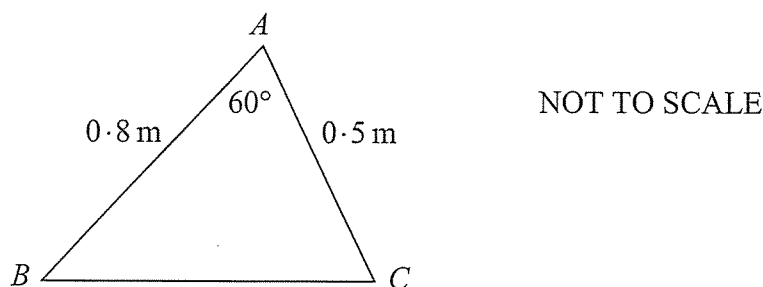
Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for questions 1-10.

1. What is the radius of the circle $x^2 + y^2 - 2y = 2$?

- (A) $\sqrt{2}$
- (B) $\sqrt{3}$
- (C) 2
- (D) 3

2.



In the diagram, ABC is a triangle in which $AB = 0.8 \text{ m}$, $AC = 0.5 \text{ m}$ and $\angle BAC = 60^\circ$.
What is the length of BC ?

- (A) 0.36 m
- (B) 0.49 m
- (C) 0.6 m
- (D) 0.7 m

3. A function f is given by the rule $f(x) = \begin{cases} (x-2)^2 + 1, & x \leq 2 \\ 4x - 7, & x > 2 \end{cases}$. What is the value of $f(f(0))$?

- (A) 5
- (B) -7
- (C) -35
- (D) 13

4. A and B are two chance events such that $P(A) = 0.45$, $P(B) = 0.3$ and $P(A|B) = 0.6$.

What is the value of $P(B|A)$?

- (A) 0.4
- (B) 0.5
- (C) 0.8
- (D) 0.9

5. The function $f(x) = \log_e x$ is transformed into the function $g(x) = \log_e 2x$.

Which of the following describes the effect of this transformation?

- (A) A horizontal translation
- (B) A vertical translation
- (C) A vertical dilation
- (D) None of the above

6. What is $\frac{d^2}{dx^2} 2^x$?

- (A) $(\log_e 2)2^x$
- (B) $(\log_e 2)4^x$
- (C) $(\log_e 2)^2 2^x$
- (D) $(\log_e 2)^2 4^x$

7. The z scores on a test are normally distributed such that $P(-N < z < N) = p$ for some constants $N > 0$ and $0 < p < 1$. What is the value of $P(z < N)$?

- (A) $\frac{1-p}{2}$
- (B) $1 - \frac{p}{2}$
- (C) $\frac{1+p}{2}$
- (D) $\frac{1+2p}{3}$

8. What is $\int (\sin^2 x + \cos^2 x + \tan^2 x) dx$?

- (A) $\tan x + c$
- (B) $\cot x + c$
- (C) $x + \sec x + c$
- (D) $x - \log_e \cos^2 x + c$

9. An item of value \$100 appreciates annually at a rate of 10% p.a. over the first n years. It then depreciates annually at a rate of 10% p.a. for the next n years. What is its final value?

- (A) \$100
- (B) \$99
- (C) $\$100(0.99)^{n^2}$
- (D) $\$100(0.99)^n$

10. What is the value of $\left(\frac{3}{5}\right)^2 \cos 2\pi + \left(\frac{3}{5}\right)^3 \cos 3\pi + \left(\frac{3}{5}\right)^4 \cos 4\pi + \left(\frac{3}{5}\right)^5 \cos 5\pi + \dots$?

- (A) $\frac{9}{40}$
- (B) $\frac{5}{8}$
- (C) $\frac{9}{10}$
- (D) $\frac{5}{2}$

Section II**90 marks****Attempt Questions 11 – 33****Allow about 2 hours and 45 minutes for this section****Instructions**

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response. If extra space is required, use the lined pages at the end of this section, carefully labelling the additional work with the appropriate question number and part.
- Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (2 marks)Find the vertex of the parabola $y = x^2 - 6x + 10$.**2**

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Question 12 (2 marks)Solve in simplest exact form the equation $\tan\left(x - \frac{\pi}{8}\right) = 1$ for $0 \leq x \leq 2\pi$.**2**

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Marks

Question 13 (2 marks)

A discrete random variable X with mean 2 has probability distribution

x	0	1	2	3
$p(x)$	a	0.1	b	0.3

2

for some constants a and b . Find the values of a and b .

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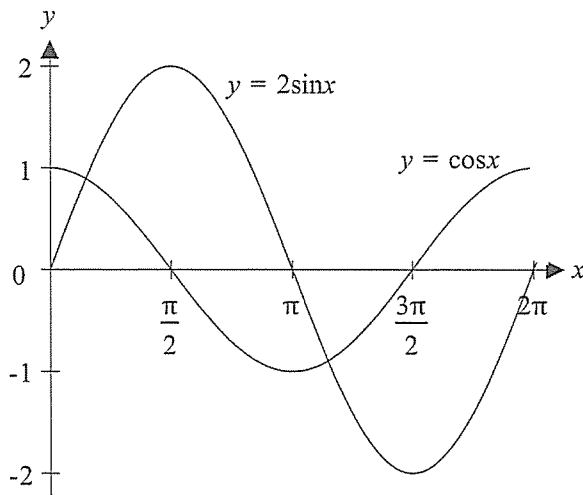
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Question 14 (2 marks)

The diagram shows the graphs of the curves $y = \cos x$ and $y = 2 \sin x$ for $0 \leq x \leq 2\pi$.

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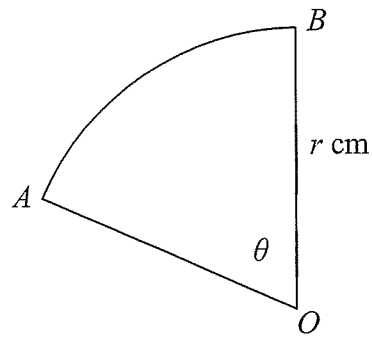
State a translation and a dilation required to transform the graph of the curve $y = \cos x$ into the graph of the curve $y = 2 \sin x$.

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Question 15 (3 marks)

NOT TO SCALE

In the diagram, AOB is a sector of a circle of radius r cm that contains an angle θ radians at the centre O of the circle. The area of the sector is 9 cm^2 and its perimeter is 5 times its arc length. Find the values of r and θ .

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Question 16 (3 marks)

Solve the equation $\log_2(1-x) + \log_2(3-x) = 3$.

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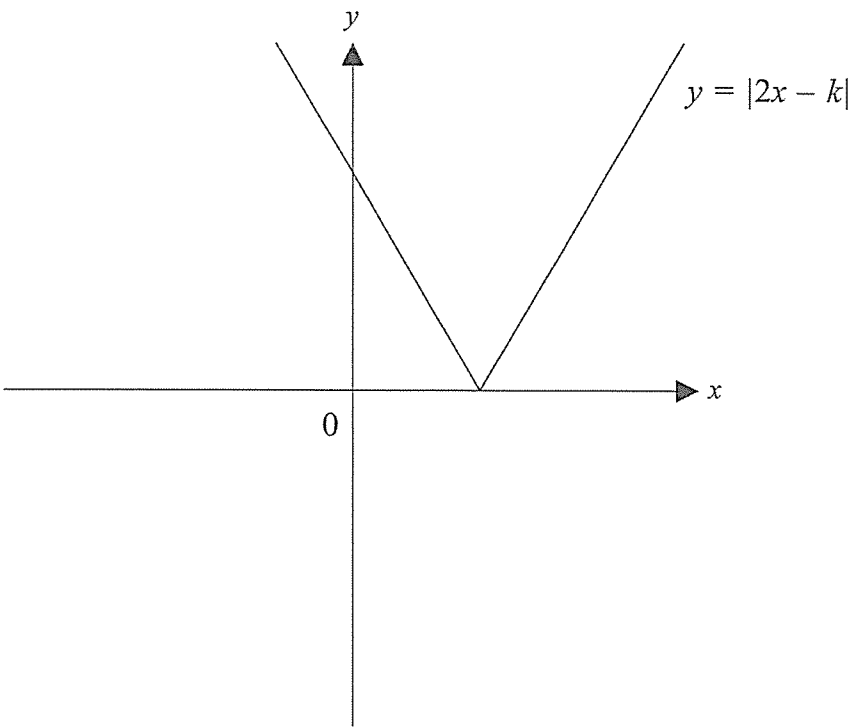
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Marks

Question 17 (3 marks)



The diagram shows the graph of the function $y = |2x - k|$ for some constant $k > 0$.

- (a) On the diagram, mark in terms of k the intercepts on the axes. 1
- (b) On the same diagram, sketch the graph of the function $y = k - |2x - k|$ showing in terms of k any intercepts on the axes and the coordinates of the sharp corner. 2

Question 18 (3 marks)

Find in simplest general form the equation of the tangent to the curve $y = e^x + \log_e(2x - 1)$ at the point where $x = 1$. 3

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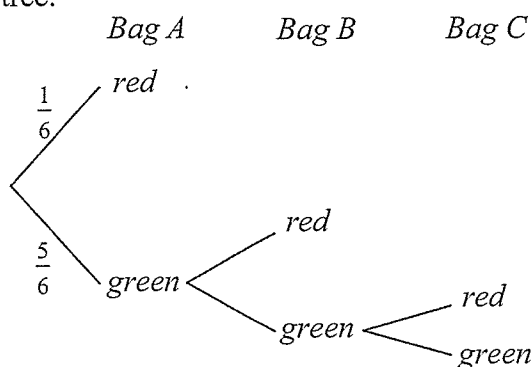
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Question 19 (4 marks)

Bag A contains 1 red ball and 5 green balls. Bag B contains 1 red ball and 3 green balls. Bag C contains 1 red ball and 2 green balls. A ball is chosen at random from Bag A. If this ball is red, no further balls are chosen, but if the ball from bag A is green, then a ball is chosen at random from bag B. If this ball is red, no further balls are chosen, but if the ball from bag B is green, then a ball is chosen at random from bag C. Let the discrete random variable X denote the number of green balls chosen.

(a) Complete the probability tree.

1



(b) Complete the probability distribution table

2

x	0	1	2	3
$p(x)$				

(c) Find in simplest fraction form the probability that at least one ball of each colour is chosen.

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Question 20 (3 marks)

If $k > 0$ is a constant find $\int_k^{2k} \frac{1}{(2x-k)^2} dx$ in simplest form in terms of k .

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Question 21 (4 marks)

The ages of the 26 million residents of Ozland are normally distributed with mean 45 years and standard deviation 11 years.

- (a) Find how many more people are older than 56 than are older than 67.

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- (b) Find correct to the nearest whole number the percentage of people younger than 56 who are younger than 34.

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Question 22 (3 marks)

If $k > 0$ is a constant, find the integer n such that $\int_k^{3k} \frac{6}{3x-k} dx = \log_e n$.

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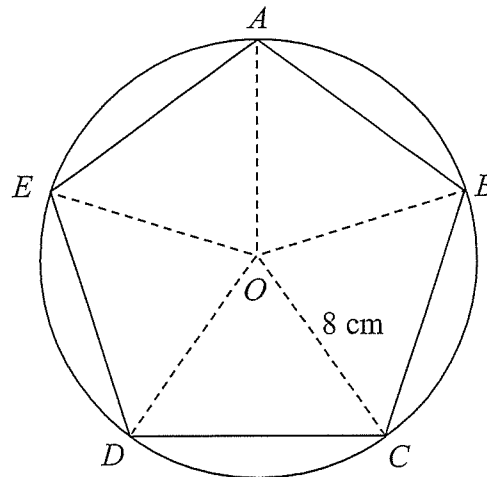
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Question 23 (4 marks)

NOT TO SCALE

In the diagram $ABCDE$ is a regular pentagon inscribed in a circle with centre O and radius 8 cm. 4
Find correct to 3 significant figures the area of $\triangle ABE$.

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Question 24 (3 marks)

If $y = \frac{\cos x}{1 + \sin x}$ show that $\frac{d^2 y}{dx^2} + y \frac{dy}{dx} = 0$.

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Question 25 (4 marks)

Each year Izzie puts 3.5% of her salary into a retirement fund. She started work at the start of 2001 with an initial salary of \$80 000 and receives an increase in salary of \$2 000 each year.

- (a) Find the year in which her yearly retirement fund contribution first exceeds \$5 000. 2

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- (b) Find the total amount that she will have put into her retirement fund when she retires at the end of 2045. 2

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Question 26 (4 marks)

- (a) If $y = \frac{x^2 + 2x}{(x+1)^2}$ find $\frac{dy}{dx}$ in the form $\frac{dy}{dx} = \frac{A}{(x+1)^n}$ for integers A and n . 3

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- (b) Find the set of values of x for which the function $y = \frac{x^2 + 2x}{(x+1)^2}$ is increasing. 1

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Question 27 (4 marks)

The table below shows the future values of an annuity of \$1 for different rates of interest and for different numbers of compounding periods. The contributions are made at the end of each time period.

Future Value Interest Factors

<i>Time Period</i>	<i>Interest Rate</i>				
	1%	2%	3%	4%	5%
1	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0200	2.0300	2.0400	2.0500
3	3.0301	3.0604	3.0909	3.1216	3.1525
4	4.0604	4.1216	4.1836	4.2465	4.3101
5	5.1010	5.2040	5.3091	5.4163	5.5256
6	6.1520	6.3081	6.4684	6.6330	6.8019
7	7.2135	7.4343	7.6625	7.8983	8.1420
8	8.2857	8.5830	8.8923	9.2142	9.5491

An annuity account is opened and contributions of \$1000 are made at the end of each quarter for 2 years at an interest rate of 4% p.a. compounding quarterly.

- (a) Find the future value of the annuity account after the last contribution is made.

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- (b) Find the single lump sum that would need to be invested at the start to achieve the same final amount at the end of the 2 years at the same compound interest rate.

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Question 28 (5 marks)

20 students sat an end-of-term Mathematics test with total mark out of 50. For each of the 20 students, the number x of Mathematics lessons missed during the term and the number y of marks lost on the test were recorded. The total number of marks lost was 320 more than the total number of Mathematics lessons missed. The correlation coefficient between lessons missed and marks lost was $r = 0.85$ and the least squares regression line of marks lost on lessons missed had equation $y = 1.2x + 14$.

- (a) Describe the nature and strength of the correlation between lessons missed and marks lost. 1

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- (b) Use the fact that the least squares regression line of marks lost on lessons missed passes through the point $(\bar{x}, \bar{y})$ to find the mean numbers of lessons missed and marks lost. 3

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- (c) Find the least squares regression line of marks gained on lessons missed. 1

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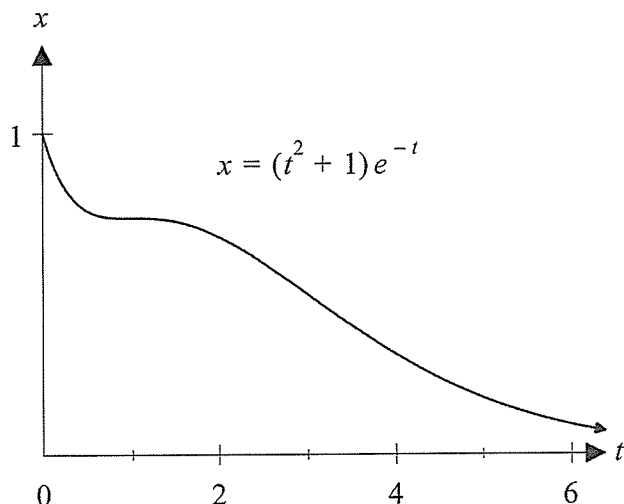
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Question 29 (5 marks)

A particle is moving in a straight line. At time t seconds it has displacement x metres from a fixed point O on the line given by $x = (t^2 + 1)e^{-t}$ and velocity $v \text{ ms}^{-1}$. The diagram shows the graph of x as a function of t .



- (a) Express v as a function of t and hence find when the particle is at rest.

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- (b) Use the features of the graph to explain whether the velocity and acceleration of the particle are positive or negative over the period of time when it is speeding up.

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- (c) Describe the limiting behaviour (speed and position) of the particle.

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Marks

Question 30 (6 marks)

Ozzie takes out a loan of \$1 000 000. Interest is charged at a rate 6% p.a. compounded monthly. At the end of each month Ozzie makes a repayment of \$ M . Ozzie pays off the loan at the end of 40 years. The amount \$ A_n owing on the loan after the n^{th} repayment is given by

$$A_n = 1\,000\,000(1.005)^n - M\left\{1 + (1.005) + (1.005)^2 + \dots + (1.005)^{n-1}\right\} \quad \text{(Do not prove this result.)}$$

- (a) Find the amount of each monthly repayment.

4

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.

- (b) Express to the nearest whole number the interest paid as a percentage of the total cost of repaying the loan.

2

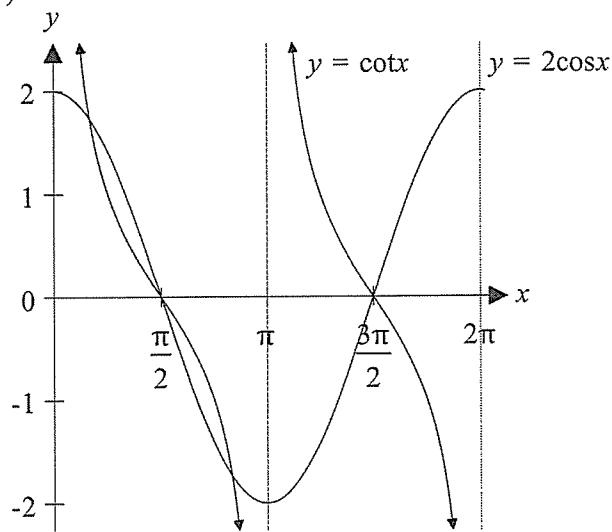
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Question 31 (7 marks)

The diagram shows the graphs of the functions $y = 2\cos x$ and $y = \cot x$ for $0 \leq x \leq 2\pi$.

- (a) Write down the periods of each of the functions $y = 2\cos x$ and $y = \cot x$.

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- (b) Show that the x coordinates of the points of intersection of the two curves are the solutions of the equation $\cos x(2\sin x - 1) = 0$ for $0 \leq x \leq 2\pi$ then solve this equation.

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Question 31 continues on page 19

Question 31 continued

- (c) Find in simplest exact form the total area of the bounded region enclosed by the two curves for $0 < x < \pi$. 3

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Question 32 (7 marks)

A continuous random variable X has a probability density function $f(x)$ given by
 $f(x) = e^{-\frac{1}{2}x}$ for $0 \leq x \leq \log_e 4$ and $f(x) = 0$ otherwise.

(a) Verify that $f(x)$ is a probability density function.

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(b) Find the cumulative distribution function $F(x)$ of X for $-\infty < x < \infty$.

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(c) Find in simplest exact form the median of X .

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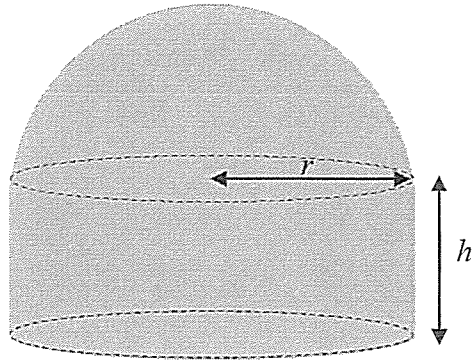
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Marks

Question 33 (7 marks)



NOT TO SCALE

A metal container of volume $0.36\pi \text{ m}^3$ is to be made in the shape of a hollow cylinder of radius r metres and height h metres, closed at the bottom and capped at the top with a hemispherical shell of the same radius. The area of metal required is $A \text{ m}^2$.

- (a) Given that a sphere of radius r has volume $\frac{4}{3}\pi r^3$ and surface area $4\pi r^2$, show that

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$$A = \frac{18\pi}{25r} + \frac{5\pi}{3}r^2.$$

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Question 33 continues on page 22

Marks

Question 33 continued

- (b) Hence find the dimensions of such a container that uses the least amount of metal.

4

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END OF PAPER

STUDENT NAME/NUMBER.....

Marks

Section II Extra writing space.

If you use this space, clearly indicate which question you are answering.

This image shows a full page of a handwriting practice worksheet. It consists of multiple sets of three horizontal dashed lines spaced evenly down the page, providing a guide for letter height and placement. The background is plain white, and there are no other markings or text present.

STUDENT NAME/NUMBER.....

Marks

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Section 1 Questions 1-10 (1 mark each)

Question	Answer	Solution	Outcomes
1	B	$x^2 + y^2 - 2y = 2$ $x^2 + (y-1)^2 = 3$ Circle with centre $(0, 1)$ and radius $\sqrt{3}$.	MA11-2
2	D	$BC^2 = 0.8^2 + 0.5^2 - 2 \times 0.8 \times 0.5 \cos 60^\circ = 0.49 \quad \therefore BC = 0.7 \text{ m}$	MA11-3
3	D	$f(0) = (0-2)^2 + 1 = 5 \quad \therefore f(f(0)) = f(5) = 4 \times 5 - 7 = 13$	MA11-2
4	A	$P(B A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A B)P(B)}{P(A)} = \frac{0.6 \times 0.3}{0.45} = 0.4$	MA11-7
5	B	$g(x) = \log_e 2x = \log_e 2 + \log_e x = \log_e 2 + f(x)$ Vertical translation up by $\log_e 2$	MA12-1
6	C	$2^x = e^{\log_e 2^x} = e^{x \log_e 2} \quad \therefore \frac{d}{dx} 2^x = (\log_e 2) e^{x \log_e 2} = (\log_e 2) 2^x$ $\therefore \frac{d^2}{dx^2} 2^x = (\log_e 2) \frac{d}{dx} 2^x = (\log_e 2)^2 2^x$	MA12-6
7	C	$P(z < N) = P(z \leq -N) + P(-N < z < N) = \frac{1}{2}(1-p) + p = \frac{1}{2}(1+p)$	MA12-8
8	A	$\int (\sin^2 x + \cos^2 x + \tan^2 x) dx = \int (1 + \tan^2 x) dx = \int \sec^2 x dx = \tan x + c$	MA12-7
9	D	$\{100(1+0.1)^n\} (1-0.1)^n = 100 \times (1.1 \times 0.9)^n = 100 \times (0.99)^n$	MA12-2
10	A	$\cos n\pi = \begin{cases} 1, & n \text{ even} \\ -1, & n \text{ odd} \end{cases} \quad \therefore \left(\frac{3}{5}\right)^n \cos n\pi = \left(\frac{3}{5}\right)^n (-1)^n = \left(-\frac{3}{5}\right)^n$ Hence require limiting sum of geometric series with $a = \left(\frac{3}{5}\right)^2$ and $r = -\frac{3}{5}$ $\therefore \sum_{n=2}^{\infty} \left(\frac{3}{5}\right)^n \cos n\pi = \frac{\left(\frac{3}{5}\right)^2}{1 - \left(-\frac{3}{5}\right)} = \frac{9}{40}$	MA12-4

Section II

Question 11

Outcomes assessed: MA11-2

Marking Guidelines

Criteria	Marks
Completes the square to state both coordinates of the vertex	2
Substantial progress eg. finds one vertex coordinate	1

Answer

$$y = x^2 - 6x + 10$$

$$y - 1 = (x - 3)^2$$

Parabola with vertex $(3, 1)$

Question 12

Outcomes assessed: MA11-3

Marking Guidelines

Criteria	Marks
States both solutions in simplest exact form	2
Substantial progress eg. finds one solution in simplest exact form	1

Answer

$$\tan\left(x - \frac{\pi}{8}\right) = 1, \quad 0 \leq x \leq 2\pi$$

$$\therefore x = \frac{3\pi}{8}, \frac{11\pi}{8}$$

$$x - \frac{\pi}{8} = \frac{\pi}{4}, \pi + \frac{\pi}{4}$$

Question 13

Outcomes assessed: MA11-7

Marking Guidelines

Criteria	Marks
Writes and solves simultaneous equations to find the values of a and b	2
Substantial progress eg. writes one equation for a and b	1

Answer

$$E(X) = 2 \Rightarrow 0.1 + 2b + 0.9 = 2 \quad \therefore b = 0.5$$

$$\sum p(x) = 1 \Rightarrow a + b = 0.6 \quad a = 0.1$$

Question 14

Outcomes assessed: AM12-1

Marking Guidelines

Criteria	Marks
States both required transformations giving appropriate detail	2
Substantial progress eg. states one of the transformations	1

Answer

Translate horizontally to the right by $\frac{\pi}{2}$ units and dilate vertically by a factor 2.

Question 15

Outcomes assessed: MA11-3

Marking Guidelines

Criteria	Marks
Writes and solves simultaneous equations to find r and θ	3
Substantial progress eg. writes simultaneous equations and finds one pronumeral	2
Some progress eg. writes one equation for r and θ	1

Answer

$$\begin{aligned} \frac{1}{2}r^2\theta &= 9 & \therefore r^2\theta &= 18 & (1) & & (2) \Rightarrow \theta &= \frac{1}{2} & \therefore r &= 6 \\ 2r + r\theta &= 5r\theta & & & 2r\theta &= r & (2) & & \text{Then } (1) \Rightarrow r^2 &= 36 & \theta &= \frac{1}{2} \end{aligned}$$

Question 16

Outcomes assessed: MA11-6

Marking Guidelines

Criteria	Marks
Writes and solves a quadratic equation for x subject to the restriction $x < 1$	3
Substantial progress eg. writes and solves a quadratic equation for x ignoring the restriction	2
Some progress eg. writes a quadratic equation for x	1

Answer

$$\begin{aligned}
 x < 1 \text{ and } \log_2(1-x)(3-x) &= 3 \\
 (1-x)(3-x) &= 8 \\
 x^2 - 4x - 5 &= 0 & \therefore x = -1 \quad (\text{Since } x < 1 \Rightarrow x \neq 5) \\
 (x-5)(x+1) &= 0
 \end{aligned}$$

Question 17

a. Outcomes assessed: MA11-1

Marking Guidelines

Criteria	Marks
States both intercepts in terms of k	1

Answer

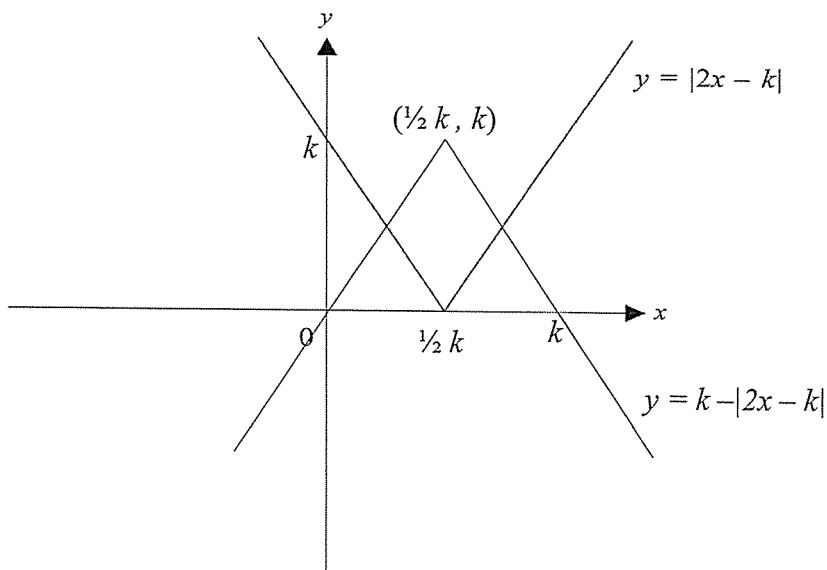
$$\begin{aligned}
 y &= |2x - k| & x = 0 \Rightarrow y &= |-k| = k & \therefore y \text{ intercept is } k \text{ and } x \text{ intercept is } \frac{1}{2}k. \\
 & & y = 0 \Rightarrow 2x &= k
 \end{aligned}$$

b. Outcomes assessed: MA12-1

Marking Guidelines

Criteria	Marks
Sketches the transformed graph showing the required features	2
Substantial progress eg. sketches graph with correct shape, position but without required detail	1

Answer



Question 18

Outcomes assessed: MA12-6

Marking Guidelines

Criteria	Marks
Uses differentiation to find gradient and hence equation of tangent in required form	3
Substantial progress eg. differentiates; evaluates y and gradient of tangent at $x=1$	2
Some progress eg. finds first derivative	1

Answer

$$y = e^x + \log_e(2x-1)$$

$$\frac{dy}{dx} = e^x + \frac{2}{2x-1}$$

$$x=1 \Rightarrow$$

$$y = e$$

$$\frac{dy}{dx} = e+2$$

Tangent at $(1, e)$ gradient $e+2$ and equation

$$y - e = (e+2)(x-1)$$

$$(e+2)x - y - 2 = 0$$

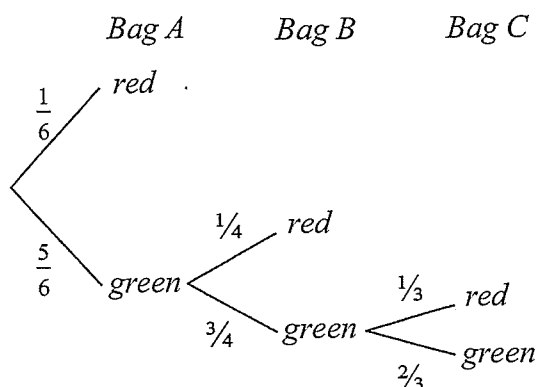
Question 19

a. Outcomes assessed: MA11-7

Marking Guidelines

Criteria	Marks
Completes probability tree	1

Answer



b. Outcomes assessed: MA11-7

Marking Guidelines

Criteria	Marks
Finds all 4 probabilities in the table	2
Substantial progress eg. finds 2 or 3 of the probabilities in the table	1

Answer

x	0	1	2	3
$p(x)$	$\frac{1}{6}$	$\frac{5}{24}$	$\frac{5}{24}$	$\frac{5}{12}$

c. Outcomes assessed: MA11-7

Marking Guidelines

Criteria	Marks
Calculates required probability	1

Answer

$$\begin{aligned}
 P(\text{at least one of each colour}) &= P(G, R) + P(G, G, R) \\
 &= \frac{5}{24} + \frac{5}{24} \\
 &= \frac{5}{12}
 \end{aligned}$$

Question 20

Outcomes assessed: MA12-7

Marking Guidelines

Criteria	Marks
Finds anti-derivative, substitutes limits in terms of k and simplifies	3
Substantial progress eg. correct procedure with one error	2
Some progress eg. finds anti-derivative	1

Answer

$$\begin{aligned}\int_k^{2k} \frac{1}{(2x-k)^2} dx &= -\frac{1}{2} \left[\frac{1}{2x-k} \right]_k^{2k} \\ &= -\frac{1}{2} \left(\frac{1}{3k} - \frac{1}{k} \right) \\ &= \frac{1}{3k}\end{aligned}$$

Question 21

a. Outcomes assessed: MA12-8

Marking Guidelines

Criteria	Marks
Uses the empirical rules to find the appropriate percentage and hence number of the population	2
Substantial progress eg. converts both numbers to z-scores, finding one appropriate percentage	1

Answer

$$\begin{aligned}X &\sim N(45, 11^2) & P(X > 56) &= P\left(Z > \frac{56-45}{11} = 1\right) = 16\% & (0.16 - 0.025) \times 26 \text{ million} \\ Z &\sim N(0, 1) & P(X > 67) &= P\left(Z > \frac{67-45}{11} = 2\right) = 2.5\% & = 3.51 \text{ million}\end{aligned}$$

b. Outcomes assessed: MA12-8

Marking Guidelines

Criteria	Marks
Uses the empirical rules to find the component percentages and deduce the required result	2
Substantial progress eg. converts numbers to z-scores, finding component percentages	1

Answer

$$\begin{aligned}P(X < 56) &= P(Z < 1) = 84\% \\ P(X < 34) &= P\left(Z < \frac{34-45}{11} = -1\right) = 16\% \\ &\left(\frac{0.16 \times 26 \text{ million}}{0.84 \times 26 \text{ million}} \times 100 \right) \% \approx 19\%\end{aligned}$$

Question 22

Outcomes assessed: MA12-7

Marking Guidelines

Criteria	Marks
Evaluates definite integral in terms of k the uses log laws to simplify, finding value of n	3
Substantial progress eg. correct procedure with one error	2
Some progress eg. finds anti-derivative	1

Answer

$$\begin{aligned}
 \int_k^{3k} \frac{6}{3x-k} dx &= 2 \left[\log_e (3x-k) \right]_k^{3k} \\
 &= 2 \{ \log_e 8k - \log_e 2k \} & \therefore n = 16 \\
 &= 2 \log_e 4 \\
 &= \log_e 16
 \end{aligned}$$

Question 23

Outcomes assessed: MA11-3

Marking Guidelines

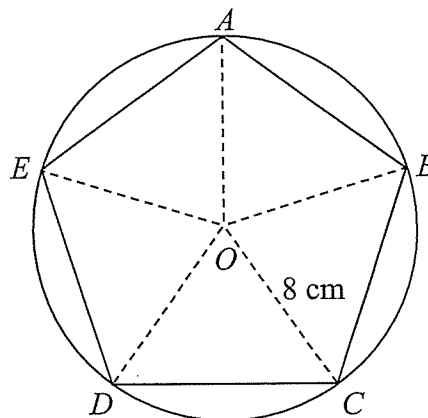
Criteria	Marks
Finds the area of each Δ congruent to ΔAOB then deduces the required area to 3 sig. fig.	4
Substantial progress eg. correct procedure with one error in calculation or accuracy	3
Moderate progress eg. finds the area of ΔAOB	2
Some progress eg. states the size of $\angle AOB$ in radians or degrees	1

Answer

The regular pentagon $ABCDE$ is divided into 5 congruent triangles each with area $\frac{1}{2} \times 8 \times 8 \times \sin \frac{2\pi}{5} = 32 \sin \frac{2\pi}{5} \text{ cm}^2$.

$$\begin{aligned}
 \text{Area } \Delta ABE &= \text{Area } \Delta AOE + \text{Area } \Delta AOB - \text{Area } \Delta EOB \\
 &= \left(2 \times 32 \sin \frac{2\pi}{5} - \frac{1}{2} \times 8 \times 8 \sin \frac{4\pi}{5} \right) \text{ cm}^2 \\
 &\approx 42.058 \text{ cm}^2
 \end{aligned}$$

Area of ΔABE is approximately 42.1 cm^2 (3 sig. fig.)



Question 24

Outcomes assessed: MA12-6

Marking Guidelines

Criteria	Marks
Finds first and second derivatives then obtains required result by substitution	3
Substantial progress eg. correct procedure with one error or lack of clarity	2
Some progress eg. finds first derivative	1

Answer

$$y = \frac{\cos x}{1 + \sin x}$$

$$\frac{dy}{dx} = \frac{-\sin x(1 + \sin x) - \cos x \cos x}{(1 + \sin x)^2}$$

$$= \frac{-\sin x - 1}{(1 + \sin x)^2}$$

$$= \frac{-1}{(1 + \sin x)}$$

$$\frac{d^2 y}{dx^2} = \frac{\cos x}{(1 + \sin x)^2}$$

$$\frac{d^2 y}{dx^2} + y \frac{dy}{dx} = \frac{\cos x}{(1 + \sin x)^2} + \frac{\cos x}{(1 + \sin x)} \times \frac{-1}{(1 + \sin x)} = 0$$

Question 25

a. Outcomes assessed: MA12-2

Marking Guidelines

Criteria	Marks
Uses expression for n^{th} term in an AP to find appropriate calendar year.	2
Substantial progress eg. finds expression for salary in n^{th} year	1

Answer

Contribution in n^{th} year is $\$T_n$ where $T_n = 0.035 \times \{80\,000 + (n-1)2\,000\} = 2\,730 + 70n$.

$$T_n > 5000 \Rightarrow 70n > 2270$$

$$n > 32.43$$

Contribution first exceeds \$5 000 in 2033.

b. Outcomes assessed: MA12-2

Marking Guidelines

Criteria	Marks
Evaluates sum of 45 terms of the AP to find required amount	2
Substantial progress eg. substitutes into expression for sum of 45 terms in an appropriate AP	1

Answer

$$\sum_{n=1}^{45} T_n = \frac{45}{2} (T_1 + T_{45}) = \frac{45}{2} (2\,800 + 5\,880) = 195\,300$$

Total amount at end of 2045 is \$195 300

Question 26

a. Outcomes assessed: MA11-5

Marking Guidelines

Criteria	Marks
Finds first derivative in required form	3
Substantial progress eg. finds first derivative but does not simplify into required form	2
Some progress eg. applies quotient rule or rearranges expression for y into convenient form	1

Answer

$$y = \frac{x^2 + 2x}{(x+1)^2} \quad \frac{dy}{dx} = 2(x+1)^{-3}$$

$$= \frac{(x+1)^2 - 1}{(x+1)^2} = \frac{2}{(x+1)^3}$$

$$= 1 - (x+1)^{-2}$$

b. Outcomes assessed: MA12-6

Marking Guidelines

Criteria	Marks
States correct inequality for x	1

Answer

$$\frac{dy}{dx} > 0 \text{ for } (x+1)^3 > 0. \text{ Function increasing for } x > -1 \text{ or } (-1, +\infty) \text{ in interval form.}$$

Question 27

a. Outcomes assessed: MA12-2

Marking Guidelines

Criteria	Marks
Selects and uses appropriate cell in table to calculate future value	2
Substantial progress eg. correct procedure with one error in cell selection or calculation	1

Answer

$$\text{Future value is } 8.2857 \times \$1000 = \$8\,285.70$$

b. Outcomes assessed: MA12-2

Marking Guidelines

Criteria	Marks
Calculates present value	2
Substantial progress eg. writes an equation or numerical expression for the present value	1

Answer

$$P(1+0.01)^8 = 8\,285.70$$

$$\$7\,651.70 \text{ (nearest cent)}$$

$$P = 8\,285.70 \div (1.01)^8 \approx 7651.7049$$

Question 28

a. Outcomes assessed: MA12-8

Marking Guidelines

Criteria	Marks
Correlation described as strong and positive	1

Answer

The value of r indicates a strong, positive linear relationship

b. Outcomes assessed: MA12-8

Marking Guidelines

Criteria	Marks
Writes and solves simultaneous equations to find the required means	3
Substantial progress eg. correct procedure with one error in solution	2
Some progress eg. writes one equation for the two means	1

Answer

$$\begin{aligned}
 20\bar{y} &= 20\bar{x} + 320 & 5\bar{y} &= 5\bar{x} + 80 \quad (1) & \bar{x} &= 10 \\
 \bar{y} &= 1.2\bar{x} + 14 & 5\bar{y} &= 6\bar{x} + 70 \quad (2) & \bar{y} &= 26 \\
 & & (2) - (1) &\Rightarrow \bar{x} - 10 = 0
 \end{aligned}$$

Mean number of lesson missed is 10, while mean number of marks lost is 26.

c. Outcomes assessed: MA12-8

Marking Guidelines

Criteria	Marks
States the appropriate least squares regression line	1

Answer

For each student, since the test is out of 50, $Marks\ gained = 50 - y$.

Hence least square regression line for marks gained on lessons lost is $Marks\ gained = 50 - (1.2x + 14)$,
ie. $Y = -1.2x + 36$ where Y is the number of marks gained on the test

Question 29

a. Outcomes assessed: MA12-6

Marking Guidelines

Criteria	Marks
Differentiates to find v as a function of t and solves $v=0$, interpreting solution	2
Substantial progress eg. finds v as a function of t by differentiation	1

Answer

$$\begin{aligned}
 v &= 2t e^{-t} + (t^2 + 1)(-e^{-t}) \\
 x &= (t^2 + 1)e^{-t} & &= -e^{-t}(t^2 - 2t + 1) & v &= 0 \text{ for } t = 1. \\
 & & &= -e^{-t}(t - 1)^2
 \end{aligned}$$

Particle is instantaneously at rest 1 second after observation starts.

Q29 (cont)

b. Outcomes assessed: MAMA12-6

Marking Guidelines

Criteria	Marks
Uses the features of the curve to deduce both v and a negative when particle speeding up	2
Substantial progress eg. explains why v is always negative when speeding up	1

Answer

Since the curve is always falling as t increases, $v < 0$ for all t except $t = 1$ when $v = 0$. Hence for particle to be speeding up, v would be becoming more negative over time so that the gradient of the graph would be negative and decreasing over time and thus the curve would be concave down.

Hence for particle to be speeding up, both velocity and acceleration would be negative.

c. Outcomes assessed: MA12-6

Marking Guidelines

Criteria	Marks
Describes limiting position and speed	1

Answer

Particle approaches O from its initial position with speed approaching zero.

Question30

a. Outcomes assessed: MA12-2

Marking Guidelines

Criteria	Marks
Sums the GP to simplify A_n then solves $A_{480} = 0$ for M to determine the monthly repayment	4
Substantial progress eg. correct procedure with one error in solution for M	3
Moderate progress eg. sums the GP and writes an equation for M using $A_{480} = 0$	2
Some progress eg. sums the GP	1

Answer

$$A_n = 1\,000\,000(1.005)^n - M\{1 + (1.005) + (1.005)^2 + \dots + (1.005)^{n-1}\}$$

$$= 1\,000\,000(1.005)^n - M \frac{(1.005)^n - 1}{1.005 - 1}$$

$$= 1\,000\,000(1.005)^n - 200M\{(1.005)^n - 1\}$$

$$A_{480} = 0 \Rightarrow 200M\{(1.005)^{480} - 1\} = 1\,000\,000(1.005)^{480}$$

$$M = \frac{5\,000(1.005)^{480}}{(1.005)^{480} - 1}$$

$$\approx 5502.136$$

Monthly repayment is \$5502.14.

Q30 (cont)

b. Outcomes assessed: MA12-2

Marking Guidelines

Criteria	Marks
Calculates required percentage	2
Substantial progress eg. finds interest paid	1

Answer

$$\left\{ \left(\frac{480M - 1\,000\,000}{480M} \right) \times 100 \right\} \% \approx 62\%$$

Question 31

a. Outcomes assessed: MA11-3

Marking Guidelines

Criteria	Marks
States periods of both functions	1

Answer

$y = 2\cos x$ has period 2π . $y = \cot x$ has period π .

b. Outcomes assessed: MA11-3

Marking Guidelines

Criteria	Marks
Produces required equation and finds all 4 solutions	3
Substantial progress eg. correct procedure with only one error or omission	2
Some progress eg. produces required equation or lists the two solutions evident from the graph	1

Answer

$$\begin{array}{lll} y = 2\cos x & 2\cos x = \cot x & \cos x = 0 \Rightarrow x = \frac{\pi}{2}, \frac{3\pi}{2} \\ y = \cot x & \text{intersect for } 0 \leq x \leq 2\pi \text{ where } 2\cos x \sin x = \cos x & \sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6} \\ & \cos x(2\sin x - 1) = 0 & \therefore x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2} \end{array}$$

c. Outcomes assessed: MA12-7

Marking Guidelines

Criteria	Marks
Evaluates appropriate definite integral to find area in simplest exact form	3
Substantial progress eg. correct procedure with one error	2
Some progress eg. writes appropriate definite integral and finds anti-derivative	1

Answer

$$\begin{aligned} A &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (2\cos x - \cot x) dx + \int_{\frac{\pi}{2}}^{\frac{5\pi}{6}} (\cot x - 2\cos x) dx \\ &= 2 \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \left(2\cos x - \frac{\cos x}{\sin x} \right) dx \quad (\text{by symmetry}) \\ &= 2 \left[2\sin x - \log_e \sin x \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}} \quad \text{Area is } (2 - 2\log_e 2) \text{ sq. units} \\ &= 2 \left\{ 2\left(1 - \frac{1}{2}\right) - \left(\log_e 1 - \log_e \frac{1}{2}\right) \right\} \\ &= 2 + 2\log_e \frac{1}{2} \\ &= 2 - 2\log_e 2 \end{aligned}$$

Question 32

a. Outcomes assessed: MA12-8

Marking Guidelines

Criteria	Marks
Establishes both conditions to verify that $f(x)$ is a probability density function	3
Substantial progress eg. correct procedure with one error or omission	2
Some progress eg. finds the anti-derivative but does not complete evaluation	1

Answer

$$\begin{aligned}
 \int_0^{\log_e 4} e^{-\frac{1}{2}x} dx &= -2 \left[e^{-\frac{1}{2}x} \right]_0^{\log_e 4} \\
 &= -2 \left(e^{-\frac{1}{2} \log_e 4} - 1 \right) \\
 &= -2 \left(e^{\log_e \frac{1}{2}} - 1 \right) \\
 &= -2 \left(\frac{1}{2} - 1 \right) \\
 &= 1
 \end{aligned}$$

$\therefore f(x) \geq 0$ for all x and $\int_{-\infty}^{+\infty} f(x) dx = 1$.
 Hence $f(x)$ is a probability density function.

b. Outcomes assessed: MA12-8

Marking Guidelines

Criteria	Marks
Finds the distribution function by integration, quoting $F(x)$ over $-\infty < x < +\infty$	2
Substantial progress eg. finds the appropriate definite integral in terms of x	1

Answer

$$\begin{aligned}
 \int_0^x e^{-\frac{1}{2}t} dt &= -2 \left[e^{-\frac{1}{2}t} \right]_0^x \\
 &= -2 \left(e^{-\frac{1}{2}x} - 1 \right) \\
 &= 2 \left(1 - e^{-\frac{1}{2}x} \right)
 \end{aligned}$$

$$\therefore F(x) = \begin{cases} 0, & x < 0 \\ 2 \left(1 - e^{-\frac{1}{2}x} \right), & 0 \leq x \leq \log_e 4 \\ 1, & x > \log_e 4 \end{cases}$$

c. Outcomes assessed: MA12-8

Marking Guidelines

Criteria	Marks
Solves $F(x) = \frac{1}{2}$ to evaluate the median in simplest exact form	2
Substantial progress eg. correct procedure with one error in simplification	1

Answer

$$\begin{aligned}
 F(x) = \frac{1}{2} &\Rightarrow 2 \left(1 - e^{-\frac{1}{2}x} \right) = \frac{1}{2} \\
 1 - e^{-\frac{1}{2}x} &= \frac{1}{4} \\
 e^{-\frac{1}{2}x} &= \frac{3}{4} \\
 -\frac{1}{2}x &= \log_e \frac{3}{4} \\
 x &= 2 \log_e \frac{4}{3}
 \end{aligned}$$

$\therefore$ median is $2 \log_e \frac{4}{3}$

Question 33

a. Outcomes assessed: MA11-1

Marking Guidelines

Criteria	Marks
Finds h in terms of r and hence A in terms of r	3
Substantial progress eg. correct procedure but one error or omission in execution	2
Some progress eg. expresses h in terms of r	1

Answer

$$\frac{2}{3}\pi r^3 + \pi r^2 h = 0.36\pi$$

$$r^2 h = \frac{36}{100} - \frac{2}{3}r^3$$

$$h = \frac{9}{25r^2} - \frac{2r}{3}$$

$$A = 2\pi r h + \pi r^2 + 2\pi r^2$$

$$= \pi \left\{ 2 \left(\frac{9}{25r} - \frac{2r^2}{3} \right) + 3r^2 \right\}$$

$$= \frac{18\pi}{25r} + \frac{5\pi}{3}r^2$$

b. Outcomes assessed: MA12-6

Marking Guidelines

Criteria	Marks
Finds r so that first derivative of A is zero, verifies minimum A then calculates h	4
Substantial progress eg. correct procedure but with one error or omission	3
Moderate progress eg. finds r so that first derivative of A is zero	2
Some progress eg. differentiates A with respect to r	1

Answer

$$A = \frac{18\pi}{25r} + \frac{5\pi}{3}r^2$$

$$\frac{dA}{dr} = \pi \left(\frac{-18}{25r^2} + \frac{10}{3}r \right)$$

$$= \frac{10\pi}{3r^2} \left(r^3 - \frac{27}{125} \right)$$

$$\frac{dA}{dr} = 0 \text{ for } r = \frac{3}{5}$$

$$\frac{dA}{dr} < 0 \text{ for } r < \frac{3}{5}$$

$$\frac{dA}{dr} > 0 \text{ for } r > \frac{3}{5}$$

A takes its minimum value for

$$r = \frac{3}{5} \text{ and } h = \frac{9}{25} \times \left(\frac{5}{3} \right)^2 - \frac{2}{3} \times \frac{3}{5} = \frac{3}{5}$$

Minimum amount of metal used when the radius and height of the cylinder are each 0.6 m.

Question	Marks	Content	Syllabus Outcomes	Targeted Performance Bands
1	1	Working with functions	MA11-2	2-3
2	1	Trigonometry and measure of angles	MA11-3	2-3
3	1	Working with functions	MA11-2	2-3
4	1	Probability and discrete probability distributions	MA11-7	3-4
5	1	Graphing techniques	MA12-1	3-4
6	1	Differential calculus	MA12-6	4-5
7	1	Random variables	MA12-8	4-5
8	1	Integral calculus	MA12-7	4-5
9	1	Modelling financial situations	MA12-2	5-6
10	1	Modelling financial situations	MA12-4	5-6
11	2	Working with functions	MA11-2	2-3
12	2	Trigonometric functions and identities	MA11-3	2-3
13	2	Probability and discrete probability distributions	MA11-7	2-3
14	2	Graphing techniques	MA12-1	2-3
15	3	Trigonometry and measure of angles	MA11-3	3-4
16	3	Logarithms and exponentials	MA11-6	4-5
17 a	1	Working with functions	MA11-1	2-3
b	2	Graphing techniques	MA12-1	4-5
18	3	Differential calculus	MA12-6	3-4
19 a	1	Probability and discrete probability distributions	MA11-7	2-3
b	2	Probability and discrete probability distributions	MA11-7	2-3
c	1	Probability and discrete probability distributions	MA11-7	3-4
20	3	Integral calculus	MA12-7	3-4
21 a	2	Random variables	MA12-8	2-3
b	2	Random variables	MA12-8	3-4
22	3	Integral calculus	MA12-7	4-5
23	4	Trigonometry and measure of angles	MA11-3	5-6
24	3	Applications of differentiation	MA12-6	4-5
25 a	2	Modelling financial situations	MA12-2	4-5
b	2	Modelling financial situations	MA12-2	4-5
26 a	3	Introduction to differentiation	MA11-5	3-4
b	1	Applications of differentiation	MA12-6	3-4
27 a	2	Modelling financial situations	MA12-2	2-3
b	2	Modelling financial situations	MA12-2	2-3
28 a	1	Descriptive statistics and bivariate data analysis	MA12-8	2-3
b	3	Descriptive statistics and bivariate data analysis	MA12-8	4-5
c	1	Descriptive statistics and bivariate data analysis	MA12-8	4-5
29 a	2	Applications of differentiation	MA12-6	5-6
b	2	Applications of differentiation	MA12-6	5-6
c	1	Applications of differentiation	MA12-6	5-6
30 a	4	Modelling financial situations	MA12-2	5-6
b	2	Modelling financial situations	MA12-2	2-3
31 a	1	Trigonometric functions and identities	MA11-3	2-3
b	3	Trigonometric functions and identities	MA11-3	4-5
c	3	Integral calculus	MA12-7	5-6
32 a	3	Random variables	MA12-8	3-4
b	2	Random variables	MA12-8	3-4
c	2	Random variables	MA12-8	3-4
33 a	3	Working with functions	MA11-1	5-6
b	4	Applications of differentiation	MA12-6	5-6